Arunesh Mishra

linkedin.com/in/aruneshm | arunesh.github.io

Summary

Hands-on engineering leader with strong background in Agentic AI, Algorithms and Distributed Systems. Looking for a leadership role in a challenging and high impact domain in systems, infrastructure and AI.

Experience

Co-founder and CTO June 2019 – Present

Picolo Labs *Mountain View, CA*

- Talkbox Project: On-device privacy friendly AI. Build Android and iOS on-device speech-to-text, translator, and text-to-speech models fine tuned for 37 foreign languages. Deployed with 2500+ operators at US Coast Guard, state and other federal agencies. Additinal features include document translation, question/answer and knowledge retrieval fully offline and on-device. Built custom fine-tuning pipeline and achieved SoTA on accuracy. Some opensourced https://huggingface.co/arun100
- Wave Project: Scalable end-end encrypted Virtual Reality and Video conferencing system. Funded by NSF through a highly competitive process. Launched with 250+ customers during the Covid lockdowns.
- Privacy friendly federated learning using Fully-Homomorphic encryption algorithms. Build Android/iOS infrastructure to jointly fine-tune models without a central cloud. Private beta launched with Gov customers.
- Semi-finalist at Dept of Defense Darpa AI Challenge Picolo Labs was awarded semi-finalist at an AI Challenge for finding software vulnerabilities using Fuzzing and LLMs. Build a competitive solution that can identify CVEs in C/C++ and Java code automatically with a high success rate.
- Raised seed round from Menlo Ventures, Village Global (Eric Schmidt)
- Awarded Research grant by the National Science Foundation (NSF), matched by state government awards.

Co-founder and CTO 2017 – 2019

Incept AI *San Mateo, CA*

- Chat-based AI assistant on Android to automatically take actions on the user's behalf
- Engineered a knowledge base for the AI assistant consisting of mobile app screens, action traces and user intents
- Launched and grew the app to a total of 100K+ users

Tech Lead, Senior Software Engineer Jan 2013 – Feb 2017

Google Inc *Mountain View, CA*

- Launched a network stack for peer-peer gaming on iOS and Android via Google Play Games SDK. Designed and implemented the network stack that maintains performance under changing network conditions and NAT topologies. Worked with the CEO of VectorUnit for RipTide demo at Google I/O to showcase the platform.
- Launched an API called Google Nearby that can transmit small amount of data securely over near-ultrasound for mobile phones. This can be used to authenticate people in same physical room and securely measure physical distance between phones to under 1 inch accuracy. Responsible for the stack on iOS and Android.Google Nearby grew from inception to 500M daily users.
- Formed a team under Marissa Mayer to build an Android app that can automatically determine if two people are in the same physical room.

Software Engineer, L4 2008 – 2013

Google Inc *Mountain View, CA*

- Launcehd Wifi-based location services for 3+ Billion users on Android and Chrome from inception. Build Wifi-RF based location clustering algorithms, traffic prediction and business occupancy prediction algorithms.
- Designed and implemented a distributed system to compute user location using Wifi and Cellular base stations. System self-learns from Android users with a GPS fix to push location to indoor-users using statistical and randomized signal propagation algorithms.
- Built cellular remap detection and localization technology, which powers Google's Mobile Maps software (1B+ downloads, most popular app on mobile).
- System is now used on Android, WinMo and Blackberry phones. Also works on browsers using HTML 5 Geolocation API, such as on Mozilla's Firefox and Chrome browser, also launched with Google's Toolbar for Microsoft's Internet Explorer.
- Algorithms serves over 3 Billion location computations per day on Android and powers Google's mobile Ads business unit. The success of the Wifi project helped grow the team to 70 Engineers (from 3).

Awards and Highlights

- Experience in Machine Learning, Agentic LLM use, Fine-tuning methods, NLP/NLU applications, design of AI Assistants, On-device privacy and security centric AI, Network protocols, Systems security, Fully homomorphic encryption systems (FHE).
- Recipient of prestigious Google's Founder award for innovation. Presidents Gold Medal at IIT.
- 52 awarded patents on algorithms, security, networking, machine learning and distributed systems. Best paper award at top ACM conferences.
- From 0 to 1: Started 3 product teams at Google including p2p networking on Android/iOS, Google Play Games and led them to a Billion+ daily users.
- Raised funding from Eric Schmidt (Google ex-CEO) for Picolo Labs. Awarded NSF Research Grant for startup innovation.

Technical Skills

Languages: Kotlin, Swift, Go, Rust, Java, Python, C/C++
Frameworks: Android/iOS SDKs, React, Node.js, Flask, JUnit, Material-UI, FastAPI
Developer Tools: Git, Docker, TravisCI, Google Cloud Platform, VS Code, Visual Studio, PyCharm, IntelliJ, Eclipse, Vibe coding tools
Libraries: pandas, NumPy, Matplotlib

Education

Ph.D. Systems Security, M.S. in Computer Security University of Maryland, College Park

B.Tech, IIT Guwahati, India